Trends in Endocrinology & Metabolism

CellPress

Feature Review

Circadian Rhythms, Sleep, and Disorders of Aging

Joanna Mattis[1] and Amita Sehgal[2,*]

Sleep–wake cycles are known to be disrupted in people with neurodegenerative disorders. These findings are now supported by data from animal models for some of these disorders, raising the question of whether the disrupted sleep/circadian regulation contributes to the loss of neural function. As circadian rhythms and sleep consolidation also break down with normal aging, changes in these may be part of what makes aging a risk factor for disorders like Alzheimer's disease (AD). Mechanisms underlying the connection between circadian/sleep dysregulation and neurodegeneration remain unclear, but several recent studies provide interesting possibilities. While mechanistic analysis is under way, it is worth considering treatment of circadian/sleep disruption as a means to alleviate symptoms of neurodegenerative disorders.

Trends

Sleep and circadian dysfunction are increasingly being associated with neurodegenerative disorders. Earlier studies with human subjects are now supported by findings in genetic models.

Circadian and sleep circuits are damaged in disorders like Huntington's and Parkinson's disease and disrupted sleep/circadian rhythms may exacerbate disease pathology. Thus, these two processes may feed back on each other.

Metabolic factors may provide a mechanistic link between circadian rhythms, sleep, and neurodegenerative disorders.

Sleep and circadian dysregulation could provide a screening tool to identify populations at risk for neurodegenerative disorders.

Could Age-Associated Changes in Circadian Rhythms and Sleep Contribute to Neurodegenerative Disorders?

Aging is associated with decreased circadian rhythmicity of behaviors including sleep. Age impacts sleep timing, duration, and consolidation, such that overall sleep decreases and also tends to be more fragmented in the elderly. Sleep disruption–- be it shorter sleep duration or poorer sleep quality – is correlated with worsened cognitive performance [1–4]. Aging brings with it a myriad of changes to the brain that may also impact cognitive performance, so it is difficult to isolate the specific effects of sleep and circadian changes. Nevertheless, there is a large and growing body of literature that links changes in the sleep and circadian systems – at the molecular, circuit, and behavioral levels – with normal aging and with diseases of aging such as neurodegenerative disease.

Aging of Central and Peripheral Circadian Systems

Introduction to the Circadian System

The molecular circadian oscillator comprises transcriptional–translational negative feedback loops [5,6]. In the major loop in mammals, the transcription factors BMAL1 and CLOCK heterodimerize during the early circadian day and induce transcription of genes with promoters containing circadian E-box elements. Among these genes are their own negative feedback repressors PERIOD (*Per*) and CRYPTOCHROME (*Cry*) (Figure 1). During the early circadian night, PER and CRY form complexes and repress BMAL1–CLOCK-mediated transcription, thus downregulating their own expression. PER and CRY degrade during the night, releasing negative regulation on BMAL1 and CLOCK and enabling the start of a new circadian day. In an interlocked loop, BMAL1–CLOCK target the expression of the nuclear receptors *rev-erbα* and *ROR*, which in turn inhibit and activate respectively the transcription of BMAL1 to produce cycling of *BMAL1* mRNA. Many other genes also have E-box or REV-ERB/ROR regulatory sequences, so the circadian cycle imposes rhythmic expression on genes involved in many aspects of cellular function [7]. These lead to tissue- and circuit-level oscillations, which ultimately generate overt rhythms of physiology and behavior.

[1]Department of Neurology, University of Pennsylvania, Philadelphia, PA, USA
[2]HHMI, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA, USA

*Correspondence:
amita@mail.med.upenn.edu (A. Sehgal).

© 2016 Elsevier Ltd. All rights reserved.

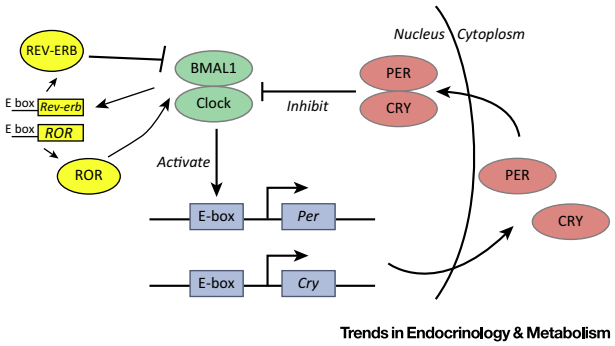


Figure 1. The Circadian Clock. BMAL1 and CLOCK are transcription factors that heterodimerize and bind to E-box element-containing promoters, including promoters for *Per* and *Cry*. PER and CRY form a complex that inhibits BMAL1–CLOCK. In an interlocked loop, BMAL1 and CLOCK target the nuclear receptors REV-ERB and ROR, which feed back to negatively and positively regulate BMAL1 transcription, respectively.

The suprachiasmatic nucleus (SCN) is the central pacemaker of the mammalian circadian system. The SCN contains a heterogeneous population of neuron types [8], including vasoactive intestinal peptide (VIP)-producing neurons in the ventral core and vasopressin (AVP) cells in the dorsal shell. Explanted SCN is capable of maintaining robust circadian rhythmicity for many days *in vitro* [9]. Many other peripheral tissues (e.g., liver, lung, skeletal muscle) are also rhythmic *in vitro* but with oscillations that are much less robust and do not persist as long [9], suggesting that the periphery contains damped oscillators that are entrained by the central SCN.

The role of the SCN as master circadian pacemaker was confirmed by seminal studies using orthotopic SCN transplantation. SCN-lesioned hamsters have permanently disrupted circadian rhythms but those rhythms can be restored by implantation of brain grafts containing fetal SCN [10]. Critical follow-up studies using mutant hamster strains with short circadian periods demonstrated that the period of circadian rhythm is determined by the genotype of the transplant SCN rather than the host [11], establishing the primacy of the SCN in the circadian system. Although the SCN signals circadian information to other brain regions via direct wiring – see below for a discussion on the circuitry relevant to sleep and arousal regions – these transplant studies are powerful evidence that SCN signaling is at least partly via secreted factors. Transplanted SCN does form some synaptic connections with the host brain post-transplantation, but it is unlikely to fully recapitulate the endogenous wiring pattern.

Changes in the Circadian System Seen in Normal Aging

Since the SCN signals both directly and indirectly to numerous brain regions, clock function and aging may be linked by pathology at the level of the SCN, SCN projections, and SCN-secreted signals. In addition, clock activity in other brain regions and in peripheral tissues may change with age.

Although rest–activity rhythms are clearly impaired with age, the core clock in the SCN appears to remain relatively robust despite some age-related changes in expression of individual clock proteins [12–14]. Data are mixed regarding whether aging impairs SCN entrainment to light [15,16]. However, it is clear that peripheral oscillators dampen with age, whether from loss of intrinsic clock function or decline in entraining signals from the SCN, and could contribute to the aging process [17]. Although changes in clock gene expression are still not causally linked to pathology associated with aging, we note that several clock targets are relevant to much of this pathology. For instance, REV-ERBα modulates the expression of genes in several metabolic and inflammatory pathways [18–21]. Also, sirtuins are involved in clock regulation and also in the control of aging [14].

Although the SCN is relatively resistant to age at the level of the molecular clock, it undergoes significant age-related degradation at the network level. The total number of SCN neurons is

unchanged with aging but elderly rats have significantly fewer vasopressinergic cells [22], a change that is likely to have consequences for downstream signaling. A greater proportion of SCN cells are silent in older animals *in vitro* [23] and SCN neurons display decreased circadian phase coherence with age [24]. This results in desynchronization [24] and decreased amplitude of electrical activity rhythms [25] at the network level. Together these are predicted to reduce the strength of output rhythms and could account for the attenuation of behavioral rhythms in older animals.

Age-associated changes in the circadian system have also been examined in *Drosophila*. As in mammals, the central brain clock is robust in old flies although the rest–activity rhythm is impaired, most likely due to deficits in output [26]. However, oscillations in body clocks are considerably dampened, most likely from loss of clock function as peripheral clocks in *Drosophila* contain their own photoreceptors and so do not rely on entraining signals from the central clock in the presence of light–dark cycles. This raises the question of whether such dampening contributes to the aging process or is merely a consequence of it. Interestingly, the *Drosophila* clock mutants *period* and *timeless* (orthologs of mammalian Per and Cry) are less sensitive to the lifespan-extending effects of dietary restriction (DR) [27]. Together with the observation that DR strengthens circadian oscillations in peripheral tissues, these findings suggest that clocks, at least those in peripheral tissues, are relevant for determination of lifespan under some conditions.

Aging of the Sleep System

Introduction to the Sleep System

Sleep and wakefulness are regulated by multiple brain regions, including the ventrolateral preoptic area (VLPO) in the anterior hypothalamus, hypocretin (orexin) neurons in the lateral hypothalamus, and the locus coeruleus (LC) in the pons (Figure 2). The VLPO contains inhibitory (galaninergic and GABAergic) neurons that are active during sleep [28] and targeted VLPO lesions result in profoundly disrupted sleep [29]. Hypocretins are neuropeptides [30] expressed in neurons that fire most during active waking and are almost silent during sleep [31]. Activation of these neurons promotes wakefulness [32,33]. Hypocretinergic neurons send dense excitatory projections to multiple nuclei including the LC [34], a noradrenergic nucleus that also promotes arousal [35]. Both the VLPO and hypocretinergic neurons send projections to brainstem regions where REM sleep is regulated [36,37]. In the 'flip–flop' model of sleep and wakefulness, the sleep-active VLPO and the wake-active monoaminergic nuclei (including

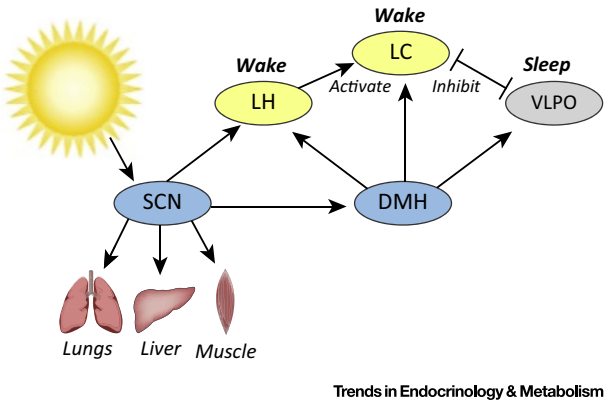


Figure 2. Sleep and Circadian Brain Circuitry in Mammals. The suprachiasmatic nucleus (SCN) is the central circadian pacemaker, entraining other brain regions as well as peripheral tissues. The SCN projects directly to the wake-promoting lateral hypothalamus (LH), which contains hypocretin neurons, and to the dorsomedial hypothalamus (DMH). The DMH projects broadly to sleep and arousal centers. Abbreviations: LC, locus coeruleus; VLPO, ventrolateral preoptic area.

the LC) mutually inhibit each other, creating rapid transition between sleep and wake states; the wake-active hypocretin neurons reinforce the arousal system and stabilize the switch [38].

Since sleep is a strongly circadian behavior, it is expected that the SCN regulates sleep and arousal brain regions. However, the SCN sends only sparse projections to hypocretin neurons [39] and no direct projections to the LC [40] or to the VLPO [41,42], suggesting a polysynaptic signaling pathway instead. Anatomic tracing studies indicate the dorsomedial hypothalamus (DMH) as a nexus of this polysynaptic circuit. The DMH is a region notable for receiving inputs from most major nuclei and areas of the hypothalamus [43] including from the SCN [43–45]. The DMH in turn projects to the LC [40], to the VLPO [42,45], and to hypocretin neurons in the lateral hypothalamus [45]. Lesions of the DMH eliminate circadian changes in LC activity [40] and reduce circadian rhythms of wake–sleep, feeding, locomotor activity, and corticosteroid secretion [45], confirming its critical role in the circuit.

Changes in the Sleep System Seen in Normal Aging

Normal aging is associated with significant changes in sleep. Elderly adults exhibit less total sleep time, poorer sleep efficiency, increased sleep latency, increased night-time awaking, and excessive daytime sleepiness with increased daytime naps [46–49]. Similar age-associated changes have also been shown in animal models spanning nonhuman primates [50], rodents [25,51–55], and flies [56].

Normal aging is also associated with changes in neural sleep circuitry; hypocretin neurons and their projections appear to be particularly altered with age. Several studies have demonstrated decreased hypothalamus hypocretin levels with aging in mice [57,58], although rhesus macaques do not display an age-related decrease in numbers of hypocretin neurons in the lateral hypothalamus [59]. A recent study also found that hypocretin expression levels decline with aging in humans [60]. Hypocretin neurons may undergo additional impairments besides changes in hypocretin expression levels; for instance, older mice reportedly have increased endoplasmic reticulum stress and dyshomeostasis in wake-active hypocretin neurons [53]. Hypocretin signaling pathways are also altered with age, with declining levels of hypocretin receptor expression [61], and with degradation of hypocretinergic projections [59,62].

Sleep and Circadian Disruptions Are Common in Neurodegenerative Disease

Disrupted sleep and circadian rhythms are common in people with neurodegenerative disease. Since these diseases disproportionately affect the elderly, who may also have other medical conditions and take multiple medications, it can be complicated to determine how much observed behavior is specifically due to the neurodegenerative disease process. Nevertheless, a large and growing body of clinical studies supports the fact that neurodegenerative diseases are associated with sleep and circadian disruption.

AD

AD is a chronic neurodegenerative disease with clinical symptoms of progressive short-term memory loss, disorientation, mood swings, loss of motivation, and behavioral changes. Pathologically it is characterized by the presence of extracellular amyloid plaques (comprising β-amyloid peptides) and by intracellular neurofibrillary tangles (comprising Tau) within the brain [63]. AD progresses through preclinical pathologic stages before the disease burden reaches the threshold for being clinically detectable [64]; sleep pathology is seen even in cognitively normal individuals with β-amyloid deposition, suggesting that the presence of β-amyloid impacts sleep architecture [65].

AD patients have sleep–wake cycle and rest–activity dysfunction [66,67]. Increased daytime sleepiness is correlated with greater functional impairment [68] and is an independent risk factor

(controlling for age) for the presence of dementia [69]. Circadian core-body temperature cycling also appears to be disrupted in AD [66,70], although AD patients do appear to retain robust rhythmic daily cortisol profiles [71,72]. A study by Cermakian and colleagues found that although patients with AD still have cycling clock gene expression in multiple brain regions [bed nucleus of the stria terminalis (BNST), cingulate cortex, and pineal gland], there is a loss of typical phase coherence within regions and of phase relationships between regions [73].

Circadian disruption in AD patients is likely to occur from disruption within clock circuits and from deficits in outputs. AD patients exhibit significant neuronal loss in the SCN [74–76] and analysis of actigraphy data and post-mortem brain tissue analysis from the same human population of AD subjects shows a correlation between circadian rhythm amplitude of motor activity and SCN neuronal loss [77]. However, flies expressing β-amyloid have circadian behavioral disruption [78] despite intact cell numbers and oscillations of the central clock, indicating additional disruption at the level of clock outputs. Consistent with this, in post-mortem analysis of brains from patients with AD, neurofibrillary degeneration in the hypothalamus was primarily observed in cortical projection neurons [79].

Parkinson's Disease (PD)

PD is a neurodegenerative disorder clinically characterized by motor symptoms including resting tremor, rigidity, and slow movement (bradykinesia); in later stages, the disease can also lead to cognitive decline and to behavioral and emotional problems. Pathologically PD is characterized by loss of midbrain dopaminergic neurons and by abnormal accumulation of the protein alpha-synuclein in intraneuronal inclusions called Lewy bodies [80]. PD also involves destruction of multiple wake-active neuronal populations [81], including a decrease in the number of hypothalamic hypocretin neurons [82,83].

Sleep disturbances are among the most morbid and prevalent non-motor symptoms of PD, with nearly two-thirds of PD patients reporting some form of sleep symptoms [84]. Multiple sleep disorders are associated with PD, including REM behavior disorder (RBD), excessive daytime somnolence, insomnia, and restless leg syndrome [84,85]. Among these, RBD, in which the skeletal muscle atonia that typically occurs during REM sleep is lost enabling complex motor activity to occur during dreams, is the most specific to PD.

Sleep abnormalities in PD are progressive over time and therefore correlate with disease duration [86]. Destruction of sleep architecture also correlates with the presence of hallucinations in PD patients [87] but not with the degree of motor impairment [86,87]. PD patients also have disrupted cortisol and melatonin regulation [72,88–90], suggesting a generalized circadian dysfunction beyond sleep–wake dysregulation. Consistent with this, peripheral (leukocyte) *BMAL1* levels are decreased in PD patients [91].

Transgenic mouse models of PD have been employed to investigate the relationship between PD and sleep pathology. Alpha-synuclein-overexpressing (ASO) transgenic mice have circadian deficits evidenced by fragmented and reduced locomotor activity that progress with age. SCN neuron firing is dampened in these mice, suggesting a weakened circadian output [92]. Another transgenic mouse line, MitoPark, exhibits circadian rhythms that are much more vulnerable to environmental insult, with profound disruption of locomotor rhythms in a constant-darkness or constant-light-exposure paradigm [93].

Observational studies following patients with known RBD demonstrate that a large majority go on to develop PD or a related disorder such as dementia with Lewy bodies [94,95]. This suggests that RBD can be seen as a prodromal phase of PD, with a mean interval of ~14 years (range 5–29 years) between RBD and clinical PD onset. This temporal delay between sleep

pathology and clinical PD onset is not well understood but may relate to findings from post-mortem pathologic analysis, which suggests that PD neurodegenerative disease often progresses undetected over many years through multiple presymptomatic phases until becoming clinically apparent [96]. Interestingly, the disease progression is often topographically predictable: pathology first begins in the brainstem and olfactory system, then progresses to include the substantia nigra, and finally involves the neocortex [96]. Consistent with this, sleep disorders such as excessive daytime somnolence [97] and RBD [94] often predate the onset of the canonical motor symptoms of PD.

Huntington's Disease (HD)

HD is an autosomal-dominant neurodegenerative disorder involving problems with mood and cognition as well as general loss of coordination and abnormal involuntary writhing movements termed chorea. The disease is caused by expansion of a CAG triplet repeat within the *Huntingtin* gene that results in a different form of the Huntingtin protein. This aberrant protein damages cells in the brain through mechanisms that, although not fully understood, include the creation of protein aggregates that form inclusion bodies within cells.

Patients with HD have progressively disordered sleep, with reduced sleep efficiency and total sleep duration [98,99]. Symptoms may include insomnia, RBD, advanced sleep phase, and reduced REM sleep [100]. Asymptomatic abnormalities in sleep architecture are detectable even early in the disease process [101]. Transgenic HD model mice (R6/2 mice) also exhibit sleep disturbances, corresponding with dysregulated expression of *mPer2* and *mBmal1* in the SCN [102]. Interestingly, the SCN from these mice demonstrates normal circadian gene expression and electrophysiological output *in vitro*, suggesting that the abnormality is due to disrupted SCN afferents as opposed to an intrinsic SCN disruption [103]. These mice have reduced VIP expression in the SCN despite preserved total SCN cell count [104]. Human patients with HD show significant loss of vasopressin, oxytocin, and hypocretin neurons [105,106].

What Are the Causal Links Between Circadian Dysfunction and Neurodegenerative Disease?

Neurodegenerative diseases correlate clinically with disruption of sleep and circadian rhythms, but correlation does not prove causation and neither does it inform the direction of causation. Does neurodegenerative disease cause sleep and circadian dysfunction or does sleep and circadian dysfunction promote neurodegeneration? Or, since these possibilities are not mutually exclusive, are both true? Answering these questions requires posing numerous additional ones. Does circadian dysfunction make the brain more vulnerable to neurodegenerative disease processes; for instance, via oxidative stress or via dysregulation of metabolism? Is sleep protective against neurodegeneration and, conversely, does lack of sleep potentiate neurodegeneration?

Circadian Dysfunction Leads to Neuronal Damage: Mechanistic Insights

Do circadian genes contribute to the onset and progression of neurodegeneration? One link is found in Presenilin-2, a protein that is involved in the cleaving of amyloid precursor protein and mutations in which are a major cause of autosomal-dominant hereditary cases of AD [107]. The *Presenilin-2* gene contains several upstream E boxes and is under circadian control, with direct gene activation by both CLOCK and BMAL1 [108]. Evidence from *Drosophila* suggests another link: SPAGHETTI (SPAG) is a regulator of the circadian kinase DOUBLETIME (DBT) and affects aggregation of Huntingtin [109]. SPAG protects DBT from degradation, stabilizing its expression [110]. Reduction of either DBT or SPAG activates the caspase DRONC and leads to DRONC-dependent Tau cleavage with increased Tau toxicity and neurodegeneration.

More generally, oxidative stress is hypothesized to be a feature shared across neurodegenerative disorders and may contribute to protein misfolding and aggregation. Multiple studies link

circadian disruption, aging, and oxidative stress. *Drosophila* with a null mutation in the *period* gene showed less resilience to oxidative stress, with increased oxidative damage and increased mortality [111]. These flies also had increased neuronal degeneration relative to age-matched controls, perhaps as a result of impaired stress defense. Double-mutant flies containing both the *period* null mutant and a mutation in *sniffer*, which on its own leads to an oxidative stress and neurodegenerative phenotype, displayed accelerated neuronal degeneration and reduced lifespan [112]. In another *Drosophila* study, the breakdown of sleep–wake cycles seen with aging was also seen with increased oxidative stress, suggesting that age-associated accumulation of oxidative damage may contribute to sleep–wake cycle disruption [56].

Several studies have investigated aging and oxidative stress in *Bmal1*$^{-/-}$ mice. These mice have a premature aging phenotype including accelerated reduction of bone and muscle mass, less subcutaneous fat, decreased hair growth, development of cataracts, and significantly reduced lifespan [113]. Reactive oxygen species (ROS) accumulation was significantly higher in *Bmal1*$^{-/-}$ mice and correlated with age-dependent pathology in specific tissues. Therefore, the authors concluded that the mechanism of the early aging in *Bmal1*$^{-/-}$ mice is their increased stress response. In support of this conclusion, the same authors later showed that supplementing these *Bmal1*$^{-/-}$ mice with the antioxidant *N*-acetyl-L-cysteine led to a partial rescue of their decreased lifespan and age-dependent pathology [114]. Musiek and colleagues found increased astrocytosis in *Bmal1*$^{-/-}$ mice relative to age-matched controls [115]. *Bmal1*$^{-/-}$ mice also showed increased markers of neuronal oxidative damage. Since mice with a brain-specific *Bmal1* deletion (*NestinCre*$^{+}$;*Bmal1*$^{f/f}$ mice) also exhibited increased astrocytosis despite grossly normal locomotor circadian rhythms and sleep–wake cycling, the authors concluded that the brain phenotype is due to increased oxidative damage as opposed to sleep–wake disturbance or systemic circadian mechanisms [115]. Given that BMAL1 is a transcriptional activator, its loss could produce phenotypes due to downregulation of target genes as opposed to loss of their cycling. Thus, it is unclear whether phenotypes of *Bmal1*$^{-/-}$ mice can be attributed to disrupted circadian function.

Metabolic Changes May Mediate Effects of Circadian/Sleep Disruption on Neurodegenerative Disorders

Metabolic disruption is correlated with neurodegenerative disorders, particularly AD. Clinically, insulin resistance is associated with increased risk of developing AD [116], and childhood obesity is not only a risk factor for diabetes but may also increase the risk of developing cognitive impairment at later life stages [117]. Further evidence comes from studies of apolipoprotein E (APOE), an important regulator of lipid metabolism expressed mainly in brain astrocytes and liver. The APOE ε4 allele impairs mitochondrial function [118], which may contribute to insulin resistance and metabolic defects [119], and is also a major risk factor for AD [120]. Young APOE ε4 carriers have dyslipidemia and reduced glucose metabolism in the same brain areas as older subjects with AD. Taken together, these data suggest that peripheral metabolic dysfunction contributes to the development of AD-associated neuropathology.

Most metabolic activity is under circadian control and loss of circadian clocks has been associated with cellular and system-wide deficits in metabolism [121,122]. Sleep loss also has profound effects on metabolism [123–125], including an increase in markers of insulin resistance. Given these findings, it is tempting to speculate that circadian/sleep disruption confers susceptibility to AD through regulation of metabolism.

Sleep Promotes Clearance of Metabolites from the Brain

Multiple studies have shown that β-amyloid levels in the brain exhibit diurnal variation. A study trending hourly measurements of CSF β-amyloid levels in human subjects revealed significant circadian patterns, with β-amyloid concentrations correlating inversely with the amount of sleep

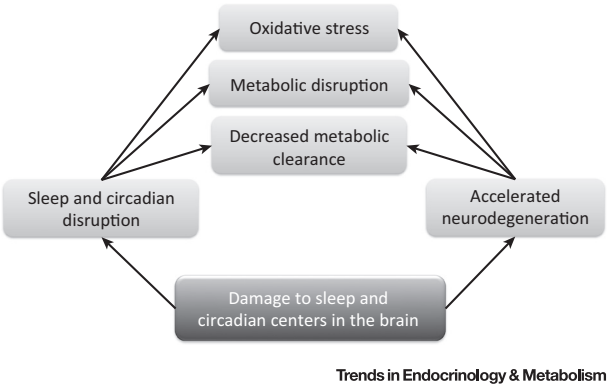


Figure 3. Possible Causal Links Between Neurodegenerative Disorders and Disruptions in Sleep and Circadian Rhythms. Sleep and circadian disruption may lead to oxidative damage, metabolic disruption, and decreased clearance of metabolites such as β-amyloid. These may accelerate neurodegeneration. Neurodegeneration may also impact brain centers that control sleep and circadian behavior.

Outstanding Questions

To what extent is there a causal relationship between sleep/circadian disruption and neurodegenerative disorders?

Do effects of circadian and sleep dysfunction on metabolism confer susceptibility to neurodegenerative disorders?

Can symptoms of neurodegenerative disorders be alleviated by improving sleep cycles?

[126]. In a rodent model, interstitial fluid levels of β-amyloid in the brain also fluctuate with the normal sleep–wake cycle [127]. The diurnal variation of β-amyloid levels suggests that β-amyloid may be cleared from the brain during sleep. Consistent with this, sleep deprivation leads to significantly elevated interstitial fluid levels of β-amyloid [128,129] and increased plaque formation [128]. Conversely, enhancing sleep reduces β-amyloid deposition [129].

Recent breakthrough work by Xie and colleagues may provide mechanistic insight into how β-amyloid is cleared from the brain's interstitial space during sleep [130]. Using *in vivo* two-photon imaging in mice, they demonstrate that sleep is associated with improved efficiency of the so-called glymphatic system (a waste clearance pathway), with a significant (>60%) increase in the volume of the interstitial space, leading to increased exchange of CSF with interstitial fluid and more efficient clearance of β-amyloid [130].

Across humans, rodents, and *Drosophila*, plaque formation leads to reduced and fragmented sleep [65,127,129]. This suggests a positive feedback loop between sleep disruption and β-amyloid deposition, which could play a key role in the pathogenesis of AD (Figure 3). It will be important to investigate whether there is a similar process linking other aggregating proteins (e.g., ∝-synuclein, Huntingtin) to sleep pathology.

Concluding Remarks and Future Directions

It is clear that circadian/sleep dysregulation constitutes a significant part of the phenotype associated with several neurodegenerative disorders, perhaps more so with some (e.g., PD) than with others. While mechanisms linking the two processes remain to be elucidated, recent studies have implicated specific proteins. Going forward, it will be important to dissect the relevant molecular and cellular pathways and also to establish causality; that is, do circadian/sleep disruptions indeed increase susceptibility to neurodegeneration and vice versa (see Outstanding Questions)? However, meanwhile it is worth considering the circadian and sleep phenotypes for diagnostic and therapeutic purposes.

Sleep and Circadian Regulation as a Therapeutic Tool

Regulation of sleep and circadian behavior in patients with neurodegenerative disease is a promising therapeutic strategy that warrants further attention. These therapies are notably low cost and noninvasive, with relatively low side-effect profiles. In pilot studies, bright-light therapy to strengthen circadian rhythms showed small but significant therapeutic benefit in PD patients [131,132]. Pharmaceutical and non-pharmaceutical therapies to treat sleep disturbances in patients with AD have shown promise [133,134]. Among patients with the apoE (*APOE*) ε4 allele, a known risk factor for developing AD, those with better sleep consolidation showed attenuated effects of genotype on developing AD, cognitive decline, and neurofibrillary tangle pathology on

autopsy [135]. Pharmacological imposition of sleep in transgenic HD mice reversed the SCN clock gene dysregulation and slowed cognitive decline based on behavioral testing [103].

Genetic Investigation into the Relationship Between Sleep and Neurodegeneration Using Human Narcolepsy

Narcolepsy is a sleep disorder characterized by excessive daytime sleepiness and by episodes of cataplexy (motor paralysis). While most cases are sporadic, some cases are familial, enabling linkage analysis [136]. Narcolepsy is thought to be caused by a loss of hypocretin-producing neurons [137], with some data supporting a broader neurodegenerative process underlying symptom onset [138]. Human narcolepsy therefore poses an opportunity to extend genetic investigation into the relationship between sleep and neurodegeneration.

Focus on Sleep Hygiene as a Public Health Issue

If promoting sleep and circadian rhythms is therapeutic, it stands to reason that the converse may also be true and disruption of sleep and circadian rhythms may have detrimental effects on health. Hamsters subjected to chronic phase advances of the light–dark cycle (i.e., experimental jet lag) exhibited impaired performance on memory tasks and were observed to have reduced hippocampal neurogenesis [139]. In humans, studies of airline employees show that chronic jet lag is associated with impaired performance on memory tasks and with temporal lobe atrophy [140,141]. If sleep disruption is re-conceptualized as a public health hazard, multiple practices – ranging from exposure to noise and light pollution to irregular shift schedules for low-wage workers – would be implicated.

Sleep and Circadian Dysregulation as a Screening Tool to Identify At-Risk Populations

A major obstacle in the treatment of neurodegenerative disease is that the disease is generally at a significantly advanced stage by the time it is diagnosed. Early (or 'prodromal') sleep manifestations of neurodegenerative disease present a tantalizing possibility for earlier detection and, potentially, more effective therapeutic intervention. Among otherwise asymptomatic elderly adults, those with high sleep fragmentation or decreased circadian activity have increased risk of developing cognitive decline or dementia [142,143]. Excessive daytime sleepiness often predates the onset of motor symptoms in PD [97]. Patients with RBD may be a useful population for research in treatments to modify PD progression and are a patient population to monitor carefully for developing symptoms of PD.

As our knowledge of sleep and circadian disruption in early disease states becomes more sophisticated, and as we develop therapeutic strategies to prevent progression of early disease, screening patients for sleep and circadian disruption – as we now routinely screen for diabetes, hypertension, or hyperlipidemia – will become an increasingly important part of primary care.

Acknowledgments

A.S. is the recipient of an Ellison Senior Scholar Award in Aging and an Investigator of the HHMI.

References

1. Miyata, S. *et al.* (2013) Poor sleep quality impairs cognitive performance in older adults. *J. Sleep Res.* 22, 535–541
2. Goel, N. *et al.* (2013) *Circadian Rhythms, Sleep Deprivation, and Human Performance.* (1st edn), Elsevier
3. Goel, N. *et al.* (2009) Neurocognitive consequences of sleep deprivation. *Semin. Neurol.* 29, 320–339
4. Lim, J. and Dinges, D.F. (2008) Sleep Deprivation and Vigilant Attention. *Ann. N. Y. Acad. Sci.* 1129, 305–322
5. Hardin, P.E. and Panda, S. (2013) Circadian timekeeping and output mechanisms in animals. *Curr. Opin. Neurobiol.* 23, 724–731
6. Zheng, X. and Sehgal, A. (2012) Speed control: cogs and gears that drive the circadian clock. *Trends Neurosci.* 35, 574–585
7. Lowrey, P.L. and Takahashi, J.S. (2004) Mammalian circadian biology: elucidating genome-wide levels of temporal organization. *Annu. Rev. Genomics Hum. Genet.* 5, 407–441
8. Honma, S. *et al.* (2012) *Suprachiasmatic Nucleus: Cellular Clocks and Networks.* (1st edn), Elsevier
9. Yamazaki, S. *et al.* (2000) Resetting central and peripheral circadian oscillators in transgenic rats. *Science* 288, 682–685
10. Lehman, M.N. *et al.* (1987) Circadian rhythmicity restored by neural transplant. Immunocytochemical characterization of the graft and its integration with the host brain. *J. Neurosci.* 7, 1626–1638
11. Ralph, M.R. *et al.* (1990) Transplanted suprachiasmatic nucleus determines circadian period. *Science* 247, 975–978

12. Kolker, D.E. *et al.* (2003) Aging alters circadian and light-induced expression of clock genes in golden hamsters. *J. Biol. Rhythms* 18, 159–169
13. Wyse, C.A. and Coogan, A.N. (2010) Impact of aging on diurnal expression patterns of CLOCK and BMAL1 in the mouse brain. *Brain Res.* 1337, 21–31
14. Chang, H.C. and Guarente, L. (2013) SIRT1 mediates central circadian control in the SCN by a mechanism that decays with aging. *Cell* 153, 1448–1460
15. Asai, M. *et al.* (2001) Circadian profile of *Per* gene mRNA expression in the suprachiasmatic nucleus, paraventricular nucleus, and pineal body of aged rats. *J. Neurosci. Res.* 66, 1133–1139
16. Davidson, A.J. *et al.* (2008) Resetting of central and peripheral circadian oscillators in aged rats. *Neurobiol. Aging* 29, 471–477
17. Yamazaki, S. *et al.* (2002) Effects of aging on central and peripheral mammalian clocks. *Proc. Natl. Acad. Sci. U.S.A.* 99, 10801–10806
18. Gibbs, J.E. *et al.* (2011) The nuclear receptor REV-ERBα mediates circadian regulation of innate immunity through selective regulation of inflammatory cytokines. *Proc. Natl. Acad. Sci. U.S.A.* 109, 582–587
19. Morioka, N. *et al.* (2016) Stimulation of nuclear receptor REV-ERBs regulates tumor necrosis factor-induced expression of proinflammatory molecules in C6 astroglial cells. *Biochem. Biophys. Res. Commun.* 469, 151–157
20. Sato, S. *et al.* (2014) A circadian clock gene, *Rev-erbα*, modulates the inflammatory function of macrophages through the negative regulation of *Ccl2* expression. *J. Immunol.* 192, 407–417
21. Duez, H. and Staels, B. (2008) *Rev-erbα* gives a time cue to metabolism. *FEBS Lett.* 582, 19–25
22. Roozendaal, B. *et al.* (1987) Changes in vasopressin cells of the rat suprachiasmatic nucleus with aging. *Brain Res.* 409, 259–264
23. Nygård, M. *et al.* (2005) Age-related changes in electrophysiological properties of the mouse suprachiasmatic nucleus *in vitro*. *Brain Res. Bull.* 65, 149–154
24. Farajnia, S. *et al.* (2012) Evidence for neuronal desynchrony in the aged suprachiasmatic nucleus clock. *J. Neurosci.* 32, 5891–5899
25. Nakamura, T.J. *et al.* (2011) Age-related decline in circadian output. *J. Neurosci.* 31, 10201–10205
26. Luo, W. *et al.* (2012) Old flies have a robust central oscillator but weaker behavioral rhythms that can be improved by genetic and environmental manipulations. *Aging Cell* 11, 428–438
27. Katewa, S.D. *et al.* (2015) Peripheral circadian clocks mediate dietary restriction-dependent changes in lifespan and fat metabolism in *Drosophila*. *Cell Metab.* 23, 143–154
28. Sherin, J.E. *et al.* (1996) Activation of ventrolateral preoptic neurons during sleep. *Science* 271, 216–219
29. Lu, J. *et al.* (2000) Effect of lesions of the ventrolateral preoptic nucleus on NREM and REM sleep. *J. Neurosci.* 20, 3830–3842
30. De Lecea, L. *et al.* (1998) The hypocretins: hypothalamus-specific peptides with neuroexcitatory activity. *Proc. Natl. Acad. Sci. U.S.A.* 95, 322–327
31. Lee, M.G. *et al.* (2005) Discharge of identified orexin/hypocretin neurons across the sleep–waking cycle. *J. Neurosci.* 25, 6716–6720
32. Adamantidis, A.R. *et al.* (2007) Neural substrates of awakening probed with optogenetic control of hypocretin neurons. *Nature* 450, 420–424
33. Carter, M.E. *et al.* (2009) Sleep homeostasis modulates hypocretin-mediated sleep-to-wake transitions. *J. Neurosci.* 29, 10939–10949
34. Peyron, C. *et al.* (1998) Neurons containing hypocretin (orexin) project to multiple neuronal systems. *J. Neurosci.* 18, 9996–10015
35. Carter, M.E. *et al.* (2010) Tuning arousal with optogenetic modulation of locus coeruleus neurons. *Nat. Neurosci.* 13, 1526–1533
36. Lu, J. *et al.* (2006) A putative flip–flop switch for control of REM sleep. *Nature* 441, 589–594
37. Weber, F. *et al.* (2015) Control of REM sleep by ventral medulla GABAergic neurons. *Nature* 526, 435–438
38. Saper, C.B. *et al.* (2005) Hypothalamic regulation of sleep and circadian rhythms. *Nature* 437, 1257–1263
39. Abrahamson, E.E. *et al.* (2001) The suprachiasmatic nucleus projects to posterior hypothalamic arousal systems. *Neuroreport* 12, 435–440
40. Aston-Jones, G. *et al.* (2001) A neural circuit for circadian regulation of arousal. *Nat. Neurosci.* 4, 732–738
41. Novak, C.M. and Nunez, A.A. (2000) A sparse projection from the suprachiasmatic nucleus to the sleep active ventrolateral preoptic area in the rat. *Neuroreport* 11, 93–96
42. Chou, T.C. *et al.* (2002) Afferents to the ventrolateral preoptic nucleus. *J. Neurosci.* 22, 977–990
43. Thompson, R.H. and Swanson, L.W. (1998) Organization of inputs to the dorsomedial nucleus of the hypothalamus: a reexamination with fluorogold and PHAL in the rat. *Brain Res. Rev.* 27, 89–118
44. Watts, a.G. and Swanson, L.W. (1987) Efferent projections of the suprachiasmatic nucleus: II. Studies using retrograde transport of fluorescent dyes and simultaneous peptide immunohistochemistry in the rat. *J. Comp. Neurol.* 258, 230–252
45. Chou, T.C. *et al.* (2003) Critical role of dorsomedial hypothalamic nucleus in a wide range of behavioral circadian rhythms. *J. Neurosci.* 23, 10691–10702
46. Redline, S. *et al.* (2004) The effects of age, sex, ethnicity, and sleep-disordered breathing on sleep architecture. *Arch. Intern. Med.* 164, 406–418
47. Moraes, W. *et al.* (2014) Effects of aging on sleep structure throughout adulthood: a population-based study. *Sleep Med.* 15, 401–409
48. Huang, Y.L. *et al.* (2002) Age-associated difference in circadian sleep–wake and rest–activity rhythms. *Physiol. Behav.* 76, 597–603
49. Carskadon, M.A. *et al.* (1982) Sleep fragmentation in the elderly: relationship to daytime sleep tendency. *Neurobiol. Aging* 3, 321–327
50. Zhdanova, I.V. *et al.* (2011) Aging of intrinsic circadian rhythms and sleep in a diurnal nonhuman primate, *Macaca mulatta*. *J. Biol. Rhythms* 26, 149–159
51. Valentinuzzi, V.S. *et al.* (1997) Effects of aging on the circadian rhythm of wheel-running activity in C57BL/6 mice. *Am. J. Physiol.* 273, R1957–R1964
52. Van Gool, W.A. *et al.* (1987) Age-related changes in circadian sleep–wakefulness rhythms in male rats isolated from time cues. *Brain Res.* 413, 384–387
53. Naidoo, N. *et al.* (2011) Endoplasmic reticulum stress in wake-active neurons progresses with aging. *Aging Cell* 10, 640–649
54. Shiromani, P.J. *et al.* (2000) Compensatory sleep response to 12 h wakefulness in young and old rats. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* 278, R125–R133
55. Zhang, Y. *et al.* (1996) Effects of aging on light-induced phase-shifting of circadian behavioral rhythms, Fos expression and CREB phosphorylation in the hamster suprachiasmatic nucleus. *Neuroscience* 70, 951–961
56. Koh, K. *et al.* (2006) A *Drosophila* model for age-associated changes in sleep:wake cycles. *Proc. Natl. Acad. Sci. U.S.A.* 103, 13843–13847
57. Sawai, N. *et al.* (2010) Developmental and aging change of orexin-A and -B immunoreactive neurons in the male rat hypothalamus. *Neurosci. Lett.* 468, 51–55
58. Brownell, S.E. and Conti, B. (2010) Age- and gender-specific changes of hypocretin immunopositive neurons in C57Bl/6 mice. *Neurosci. Lett.* 472, 29–32
59. Downs, J.L. *et al.* (2007) Orexin neuronal changes in the locus coeruleus of the aging rhesus macaque. *Neurobiol. Aging* 28, 1286–1295
60. Hunt, N.J. *et al.* (2015) Changes in orexin (hypocretin) neuronal expression with normal aging in the human hypothalamus. *Neurobiol. Aging* 36, 292–300
61. Terao, A. *et al.* (2002) Age-related decline in hypocretin (orexin) receptor 2 messenger RNA levels in the mouse brain. *Neurosci. Lett.* 332, 190–194

62. Zhang, J-H. *et al.* (2002) Age-related changes in hypocretin (orexin) immunoreactivity in the cat brainstem. *Brain Res.* 930, 206–211
63. Bloom, G.S. (2014) Amyloid-β and Tau. *JAMA Neurol.* 71, 505
64. Braak, H. and Braak, E. (1995) Staging of Alzheimer's disease-related neurofibrillary changes. *Neurobiol. Aging* 16, 271–278
65. Ju, Y-E.S. *et al.* (2013) Sleep quality and preclinical Alzheimer disease. *JAMA Neurol.* 70, 587–593
66. Satlin, A. *et al.* (1995) Circadian locomotor-activity and core-body temperature rhythms in Alzheimer's-disease. *Neurobiol. Aging* 16, 765–771
67. Van Someren, E.J.W. *et al.* (1996) Circadian rest–activity rhythm disturbances in Alzheimer's disease. *Biol. Psychiatry* 40, 259–270
68. Lee, J.H. *et al.* (2007) Daytime sleepiness and functional impairment in Alzheimer disease. *Am. J. Geriatr. Psychiatry* 15, 620–626
69. Merlino, G. *et al.* (2010) Daytime sleepiness is associated with dementia and cognitive decline in older Italian adults: a population-based study. *Sleep Med.* 11, 372–377
70. Lee, J.H. *et al.* (2004) Twenty-four-hour rhythms of sleep–wake cycle and temperature in Alzheimer's disease. *J. Neuropsychiatry* 6, 192–198
71. Hatfield, C.F. *et al.* (2004) Disrupted daily activity/rest cycles in relation to daily cortisol rhythms of home-dwelling patients with early Alzheimer's dementia. *Brain* 127, 1061–1074
72. Hartmann, A. *et al.* (1997) Twenty-four hour cortisol release profiles in patients with Alzheimer's and Parkinson's disease compared to normal controls: ultradian secretory pulsatility and diurnal variation. *Neurobiol. Aging* 18, 285–289
73. Cermakian, N. *et al.* (2011) Circadian clock gene expression in brain regions of Alzheimer's disease patients and control subjects. *J. Biol. Rhythms* 26, 160–170
74. Stopa, E.G. *et al.* (1999) Pathologic evaluation of the human suprachiasmatic nucleus in severe dementia. *J. Neuropathol. Exp. Neurol.* 58, 29–39
75. Zhou, J.N. *et al.* (1995) VIP neurons in the human SCN in relation to sex, age, and Alzheimer's-disease. *Neurobiol. Aging* 16, 571–576
76. Swaab, D.F. *et al.* (1985) The suprachiasmatic nucleus of the human brain in relation to sex, age and senile dementia. *Brain Res.* 342, 37–44
77. Wang, J.L. *et al.* (2015) Suprachiasmatic neuron numbers and rest–activity circadian rhythms in older humans. *Ann. Neurol.* 78, 317–322
78. Chen, K-F. *et al.* (2014) The central molecular clock is robust in the face of behavioural arrhythmia in a *Drosophila* model of Alzheimer's disease. *Dis. Model. Mech.* 7, 445–458
79. Saper, C.B. and German, D.C. (1987) Hypothalamic pathology in Alzheimer's disease. *Neurosci. Lett.* 74, 364–370
80. Schulz-Schaeffer, W.J. (2010) The synaptic pathology of α-synuclein aggregation in dementia with Lewy bodies, Parkinson's disease and Parkinson's disease dementia. *Acta Neuropathol.* 120, 131–143
81. Stern, A.L. and Naidoo, N. (2015) Wake-active neurons across aging and neurodegeneration: a potential role for sleep disturbances in promoting disease. *Springerplus* 4, 25
82. Fronczek, R. *et al.* (2007) Hypocretin (orexin) loss in Parkinson's disease. *Brain* 130, 1577–1585
83. Thannickal, T.C. *et al.* (2007) Hypocretin (orexin) cell loss in Parkinson's disease. *Brain* 130, 1586–1595
84. Barone, P. *et al.* (2009) The PRIAMO study: a multicenter assessment of nonmotor symptoms and their impact on quality of life in Parkinson's disease. *Mov. Disord.* 24, 1641–1649
85. Chaudhuri, K.R. *et al.* (2006) International multicenter pilot study of the first comprehensive self-completed nonmotor symptoms questionnaire for Parkinson's disease: the NMSQuest study. *Mov. Disord.* 21, 916–923
86. Diederich, N. *et al.* (2005) Progressive sleep "destructuring" in Parkinson's disease. A polysomnographic study in 46 patients. *Sleep Med.* 6, 313–318
87. Whitehead, D.L. *et al.* (2008) Circadian rest–activity rhythm is altered in Parkinson's disease patients with hallucinations. *Mov. Disord.* 23, 1137–1145
88. Breen, D.P. *et al.* (2014) Sleep and circadian rhythm regulation in early Parkinson disease. *JAMA Neurol.* 71, 589–595
89. Bordet, R. *et al.* (2003) Study of circadian melatonin secretion pattern at different stages of Parkinson's disease. *Clin. Neuropharmacol.* 26, 65–72
90. Videnovic, A. *et al.* (2014) Circadian melatonin rhythm and excessive daytime sleepiness in Parkinson's disease. *JAMA Neurol.* 71, 463–469
91. Cai, Y. *et al.* (2010) Expression of clock genes *Per1* and *Bmal1* in total leukocytes in health and Parkinson's disease. *Eur. J. Neurol.* 17, 550–554
92. Kudo, T. *et al.* (2011) Circadian dysfunction in a mouse model of Parkinson's disease. *Exp. Neurol.* 232, 66–75
93. Fifel, K. and Cooper, H.M. (2014) Loss of dopamine disrupts circadian rhythms in a mouse model of Parkinson's disease. *Neurobiol. Dis.* 71, 359–369
94. Iranzo, A. *et al.* (2013) Neurodegenerative disease status and post-mortem pathology in idiopathic rapid-eye-movement sleep behaviour disorder: an observational cohort study. *Lancet Neurol.* 12, 443–453
95. Schenck, C.H. *et al.* (2013) Delayed emergence of a Parkinsonian disorder or dementia in 81% of older men initially diagnosed with idiopathic rapid eye movement sleep behavior disorder: a 16-year update on a previously reported series. *Sleep Med.* 14, 744–748
96. Braak, H. *et al.* (2004) Stages in the development of Parkinson's disease-related pathology. *Cell Tissue Res.* 318, 121–134
97. Abbott, R.D. *et al.* (2005) Excessive daytime sleepiness and subsequent development of Parkinson disease. *Neurology* 65, 1442–1446
98. Goodman, A.O.G. and Barker, R.A. (2010) How vital is sleep in Huntington's disease? *J. Neurol.* 257, 882–897
99. Aziz, N.A. *et al.* (2010) Sleep and circadian rhythm alterations correlate with depression and cognitive impairment in Huntington's disease. *Parkinsonism Relat. Disord.* 16, 345–350
100. Arnulf, I. *et al.* (2008) Rapid eye movement sleep disturbances in Huntington disease. *Arch. Neurol.* 65, 482–488
101. Goodman, A.O.G. *et al.* (2011) Asymptomatic sleep abnormalities are a common early feature in patients with Huntington's disease. *Curr. Neurol. Neurosci. Rep.* 11, 211–217
102. Morton, A.J. *et al.* (2005) Disintegration of the sleep–wake cycle and circadian timing in Huntington's disease. *J. Neurosci.* 25, 157–163
103. Pallier, P.N. *et al.* (2007) Pharmacological imposition of sleep slows cognitive decline and reverses dysregulation of circadian gene expression in a transgenic mouse model of Huntington's disease. *J. Neurosci.* 27, 7869–7878
104. Fahrenkrug, J. *et al.* (2007) Decreased VIP and VPAC2 receptor expression in the biological clock of the R6/2 Huntington's disease mouse. *J. Mol. Neurosci.* 31, 139–148
105. Gabery, S. *et al.* (2010) Changes in key hypothalamic neuropeptide populations in Huntington disease revealed by neuropathological analyses. *Acta Neuropathol.* 120, 777–788
106. Aziz, A. *et al.* (2008) Hypocretin and melanin-concentrating hormone in patients with Huntington disease. *Brain Pathol.* 18, 474–483
107. Levy-lahad, A.E. *et al.* (2015) Candidate gene for the chromosome 1 familial Alzheimer's disease locus. *Science* 269, 973–977
108. Bélanger, V. *et al.* (2006) The circadian regulation of Presenilin-2 gene expression. *Chronobiol. Int.* 23, 747–766
109. Zhang, S. *et al.* (2010) A genomewide RNA interference screen for modifiers of aggregates formation by mutant Huntingtin in *Drosophila*. *Genetics* 184, 1165–1179
110. Means, J.C. *et al.* (2015) *Drosophila* Spaghetti and Doubletime link the circadian clock and light to caspases, apoptosis and tauopathy. *PLoS Genet.* 11, e1005171
111. Krishnan, N. *et al.* (2009) The circadian clock gene period extends healthspan in aging *Drosophila melanogaster*. *Aging (Albany NY)* 1, 937–948

112. Krishnan, N. *et al.* (2012) Loss of circadian clock accelerates aging in neurodegeneration-prone mutants. *Neurobiol. Dis.* 45, 1129–1135

113. Kondratov, R.V. *et al.* (2006) Early aging and age-related pathologies in mice deficient in BMAL1, the core component of the circadian clock. *Genes Dev.* 20, 1868–1873

114. Kondratov, R.V. *et al.* (2009) Antioxidant N-acetyl-L-cysteine ameliorates symptoms of premature aging associated with the deficiency of the circadian protein BMAL1. *Aging (Albany NY)* 1, 979–987

115. Musiek, E.S. *et al.* (2013) Circadian clock proteins regulate neuronal redox homeostasis and neurodegeneration. *J. Clin. Invest.* 123, 5389–5400

116. Willette, A.A. *et al.* (2015) Association of insulin resistance with cerebral glucose uptake in late middle-aged adults at risk for Alzheimer disease. *JAMA Neurol.* 53792, 1013–1020

117. Luciano, R. *et al.* (2015) Biomarkers of Alzheimer disease, insulin resistance, and obesity in childhood. *Pediatrics* 135, 1074–1081

118. Huang, Y. and Mahley, R.W. (2014) Apolipoprotein E: structure and function in lipid metabolism, neurobiology, and Alzheimer's diseases. *Neurobiol. Dis.* 72, 3–12

119. Szendroedi, J. *et al.* (2011) The role of mitochondria in insulin resistance and type 2 diabetes mellitus. *Nat. Rev. Endocrinol.* 8, 92–103

120. Lane-Donovan, C. *et al.* (2014) More than cholesterol transporters: lipoprotein receptors in CNS function and neurodegeneration. *Neuron* 83, 771–787

121. Dibner, C. and Schibler, U. (2015) Circadian timing of metabolism in animal models and humans. *J. Intern. Med.* 277, 513–527

122. Maury, E. *et al.* (2014) Circadian disruption in the pathogenesis of metabolic syndrome. *Diabetes Metab.* 40, 338–346

123. Weljie, A.M. *et al.* (2015) Oxalic acid and diacylglycerol 36:3 are cross-species markers of sleep debt. *Proc. Natl. Acad. Sci. U.S.A.* 112, 2569–2574

124. Davies, S.K. *et al.* (2014) Effect of sleep deprivation on the human metabolome. *Proc. Natl. Acad. Sci. U.S.A.* 111, 10761–10766

125. Taheri, S. *et al.* (2004) Short sleep duration is associated with reduced leptin, elevated ghrelin, and increased body mass index. *PLoS Med.* 1, e62

126. Huang, Y. *et al.* (2012) Effects of age and amyloid deposition on Aβ dynamics in the human central nervous system. *Arch. Neurol.* 69, 51–58

127. Roh, J.H. *et al.* (2012) Disruption of the sleep–wake cycle and diurnal fluctuation of β-amyloid in mice with Alzheimer's disease pathology. *Sci. Transl. Med.* 4, 150ra122

128. Kang, J-E. *et al.* (2009) Amyloid-β dynamics are regulated by orexin and the sleep–wake cycle. *Science* 326, 1005–1007

129. Tabuchi, M. *et al.* (2015) Sleep Interacts with Aβ to modulate intrinsic neuronal excitability. *Curr. Biol.* 25, 1–11

130. Xie, L. *et al.* (2013) Sleep drives metabolite clearance from the adult brain. *Science* 342, 373–377

131. Rutten, S. *et al.* (2012) Bright light therapy in Parkinson's disease: an overview of the background and evidence. *Parkinsons Dis.* 2012, 1–9

132. Paus, S. *et al.* (2007) Bright light therapy in Parkinson's disease: a pilot study. *Mov. Disord.* 22, 1495–1498

133. Wu, Y-H. and Swaab, D.F. (2007) Disturbance and strategies for reactivation of the circadian rhythm system in aging and Alzheimer's disease. *Sleep Med.* 8, 623–636

134. Peter-Derex, L. *et al.* (2015) Sleep and Alzheimer's disease. *Sleep Med. Rev.* 19, 29–38

135. Lim, A.S.P. *et al.* (2013) Sleep modifies the relationship of APOE to the risk of Alzheimer disease and neurofibrillary tangle pathology. *JAMA Neurol.* 70, 1544–1551

136. Hor, H. *et al.* (2011) A missense mutation in myelin oligodendrocyte glycoprotein as a cause of familial narcolepsy with cataplexy. *Am. J. Hum. Genet.* 89, 474–479

137. Crocker, A. *et al.* (2005) Concomitant loss of dynorphin, NARP, and orexin in narcolepsy. *Neurology* 65, 1184–1188

138. Siegel, J.M. *et al.* (1999) Neuronal degeneration in canine narcolepsy. *J. Neurosci.* 19, 248–257

139. Gibson, E.M. *et al.* (2010) Experimental "jet lag" inhibits adult neurogenesis and produces long-term cognitive deficits in female hamsters. *PLoS ONE* 5, e15267

140. Cho, K. *et al.* (2000) Chronic jet lag produces cognitive deficits. *J. Neurosci.* 20, RC66

141. Cho, K. (2001) Chronic "jet lag" produces temporal lobe atrophy and spatial cognitive deficits. *Nat. Neurosci.* 4, 567–568

142. Lim, A.S.P. *et al.* (2013) Sleep fragmentation and the risk of incident Alzheimer's disease and cognitive decline in older persons. *Sleep* 36, 1027–1032

143. Tranah, G.J. *et al.* (2011) Circadian activity rhythms and risk of incident dementia and mild cognitive impairment in older women. *Ann. Neurol.* 70, 722–732